

Review

Optimal Arrangement Strategy of IoT Sensors in Urban Drainage Networks: A Review

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Abstract: The Urban Drainage Network (UDN) is a type of underground municipal infrastructure responsible for transporting sewage and rainwater. To keep abreast of the hydraulic and water quality conditions of the pipes and to detect problems such as pipe clogging, pollution and leakage, real-time monitoring sensors have been widely adopted, accomplished with the development of IoT technologies. However, the intricate topology and numerous nodes of drainage pipes complicate IoT sensor placement strategies, especially in the selection of sensors and the location of monitoring points. This review examines application cases of IoT sensors in UDNs and some other hydraulic networks, evaluating the characteristics and applicability of various optimal placement methods and theories. A general framework was proposed applicable to the optimal placement of IoT sensors in the UDN, including object classification–method selection–quantitative evaluation. Currently, the quantitative evaluation of monitoring schemes lacks a systematic process, and existing layout methods may not be optimal. Future research can explore dynamic optimization strategies through phased deployment and feedback iteration, which can enhance the accuracy and objectivity of sensor layout design and evaluation.

Keywords: urban drainage network; Internet of Things; optimal arrangement strategy; monitoring points



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1. Introduction

The Urban Drainage Network (UDN) is a type of underground municipal infrastructure for transferring and transporting rainwater and sewage. Owing to complex operating conditions, their inspection and maintenance are difficult [1]. Monitoring the operational status of pipeline networks using Internet of Things (IoT) sensors can facilitate the identification of problems such as blockage, pollution, and leakage, owing to factors such as pipeline failure. This is critical to the safe and stable operation of drainage networks. In a UDN, the parameters that are monitored by sensors include water quality and hydraulic parameters. By monitoring the composition of sewage [2–4], low water quality and illegal discharge points can be identified in pipelines, and corresponding measures can be implemented to reduce the operational burden of sewage treatment plants during pollution events. Moreover, by monitoring hydraulic parameters such as the flow rate and water level, the flow state inside a pipeline can be evaluated [5,6], and abnormal events, such as blockage, external water infiltration, backflow, and overflow [7–9], can be analyzed to guide the operation and maintenance of the network.

Traditional drainage network operation and maintenance rely heavily on regular field surveys [10,11] or sewage and sediment sampling [3,12,13] to obtain network operational data.

Currently, high-precision sensors such as flow, level, and current meters are widely utilized in important nodes of UDNs. However, owing to several limitations, including their data temporal resolution, the amount of data collected, and the spatial distribution of sampling points, it is difficult to adequately assess the real-time operational status of the drainage network using regular manual sampling or empirical judgment. Consequently, there is a lag in accident monitoring such as illegal discharge and flood overflow. Thus, it is impossible to reflect real-time hydraulic and water quality status in the pipeline network.

The advancement of IoT technology and the iterative improvement in sensors have led to the development of IoT sensors that facilitate real-time data transmission for safety and health monitoring of structures such as bridges [14] and dams [15], as well as environmental monitoring such as air quality [16]. They are also widely used in the monitoring of hydraulic parameters in water supply projects [17–19], surface rivers [20], groundwater [21], and rainfall [22]. Recently, IoT sensors have also been widely used in the monitoring of water quality hydraulic parameters in UDNs. The application of IoT in UDN monitoring is shown in Figure 1.

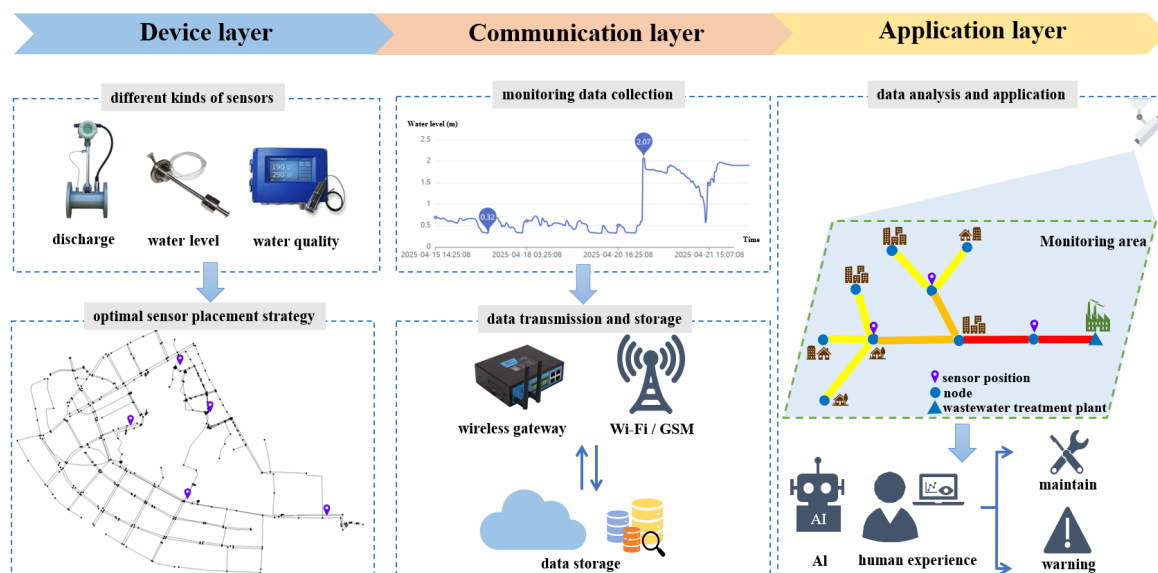


Figure 1. Demonstrates the application of IoT in drainage network monitoring.

However, the complexity of sewage composition and variable flow patterns in drainage pipes pose great challenges to the application of IoT monitoring devices and communication technologies. In addition, drainage networks have a complex pipeline network topology, consisting of pipelines at different levels, such as main drains, main sewers, branch pipelines, and connecting pipelines. Therefore, the application of IoT sensors, especially the selection of monitoring locations, is difficult. Numerous studies have been conducted on the optimized layout of sensors. A universal layout strategy has not been established yet. This investigation focused on the monitoring needs and structural characteristics of drainage pipelines, proposed a general framework applicable to the optimal layout of IoT sensors for UDNs and classified the monitored parameters at different nodes, and compared the advantages, disadvantages, and application scopes of different types of sensor layout optimization methods.

2. Literature Sources

In order to systematize the existing research results and support the framework construction, this study first clarifies the strategy and screening criteria of the literature search. The following methodology was used to search and identify relevant literature:

First, we categorized the keywords into four different dimensions according to time, type of network, technology, and topic, as shown in Table 1. Time represents the research discussed in this study, mainly up to 31 December 2024; the network types are mainly combined and separated drainage networks. In the other three dimensions, we considered terms and expressions similar to the keyword “optimal placement of IoT sensors in urban drainage networks”, and searched the Web of Science database by combining the terms of “technology” and “network”; connecting them with “and” yielded a total of 189 papers. Next, a further review in conjunction with the abstracts of the literature excluded those that were not relevant to the topic of this study. In order to ensure the comprehensiveness of the research methodology, we referenced and drew on the application of sensors in other similar hydraulic projects to provide a more robust scientific overview and suggestions for the monitoring of drainage networks. Finally, a total of 96 studies were included for discussion in this review.

Table 1. Different dimensional keywords used for literature search.

Time	Technology	Network	Theme
As of 31 December 2024	IoT/sensor online/real time monitoring/continuous measurement	Sewer/sewerage/wastewater/drainage urban networks/system	Optimal/optimization placement/site/position

As can be seen from the buzzwords and high-frequency words related to UDN monitoring in Figure 2, the current monitoring work of UDNs mainly focuses on some water quality indicators (concentration of a certain marker in the drainage network) due to the wide range of water quality characterization parameters, which can be used to identify pollution sources, analyze the trend of pollutant propagation and dispersion, and assess the water environment to facilitate managers to take timely and effective measures. The other part of the monitoring effort focuses on hydraulic characterization parameters such as flow and water level, which are mainly used to study overflow, flooding, leakage, and inflow/infiltration problems in the UDN.

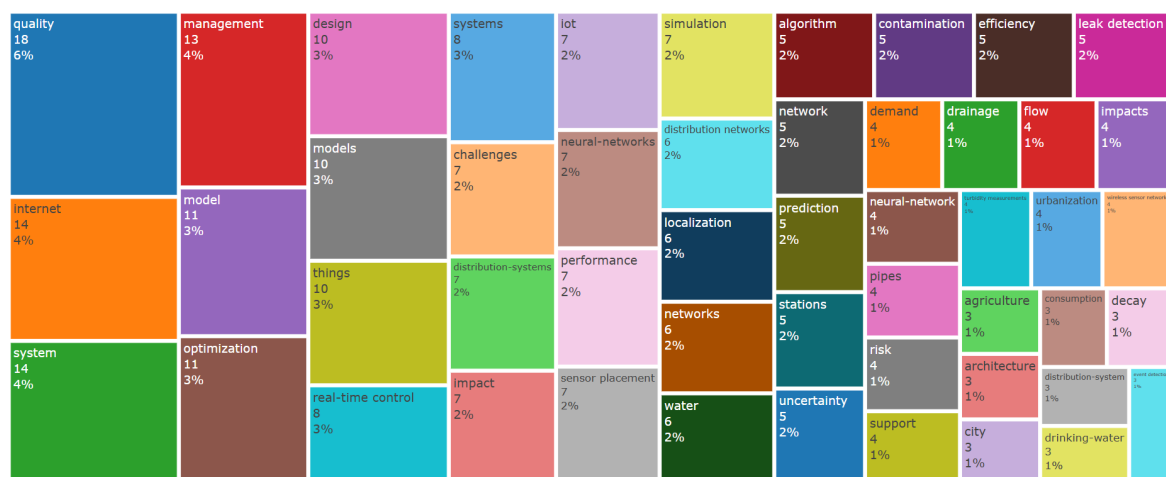


Figure 2. Hot and high-frequency words related to the monitoring of UDN.

In the process of the literature research, we found that the research and application of IoT sensors on UDN show a rising trend year by year, and these articles mainly focus on the stage of analyzing and applying the collected data on drainage networks. Nevertheless, the importance of the site selection of the monitoring point in the early stage for the whole research should not be ignored, but relatively few studies have been conducted on this key

link. The selection of monitoring points is decisive for the effective collection, processing and analysis of subsequent data, but the current attention and research investment in this area are insufficient in the academic community, and this area is in urgent need of further in-depth research and study. This study presents a summary and classification according to the optimization theories and methods used in the study and discusses the optimal arrangement of IoT sensors in drainage networks, with a view to providing scientific guidance for the siting of monitoring points in UDNs.

3. Arrangement of IoT Sensors in the Drainage Network

In contrast with the relatively mature methods for optimizing the layout of monitoring points in surface water and water supply networks, there is still a lack of comprehensive guidance for optimizing the placement of IoT sensors in drainage networks. The optimized layout of sensors in drainage networks is similar to the selection of monitoring points in other applications, but there are significant differences in topology, flow characteristics, and water quality monitoring: ① drainage networks are usually simplified as tree-like structures [13,23], with inspection wells and connections represented by nodes [12]; ② the internal environment of pipelines is complex, usually exhibiting alternating open and full flow states; ③ the rainwater and sewage transported inside the pipeline are complex in composition and corrosive. These characteristics account for the difference in the design and layout of IoT sensors in drainage networks compared to other applications.

Sensors should be arranged to provide as much valid data as possible on the targets to be monitored, which is the first step in information analysis [24], planning and management [25–27], or the adoption of appropriate measures [28]. The accuracy of decision making depends more on the precision and reliability of data, rather than the volume of the data. Poor-quality datasets may lead to decision-making biases [29,30]. It has been shown that an increase in the number of sensors leads to a sharp decline in the return on scale [2,23]. In Figure 3, we explicitly present a general process applicable to the optimal placement of IoT sensors in UDNs, as well as a description of the content and framework of this review.

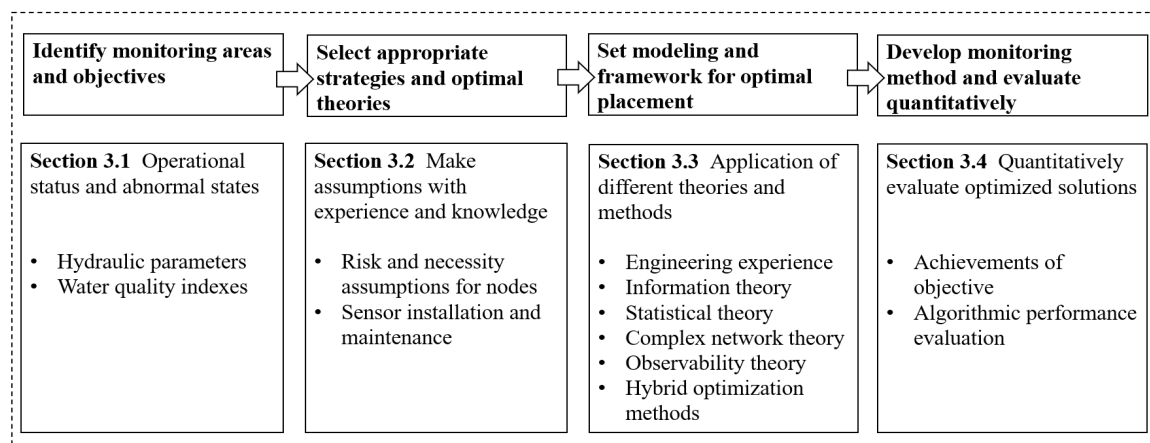


Figure 3. Process of optimizing the placement of IoT sensors in UDN.

Firstly, the study area and monitoring objectives should be identified for the selection of indicator parameters and sensor types. Topological information and data on the drainage network should be collected and organized. Based on various optimization theories and methods, the optimization model and framework were established by incorporating prior knowledge and engineering experience. The monitoring plan was then generated and quantitatively evaluated. The operational status of the drainage network can then be

obtained using the IoT monitoring data. Thus, the drainage network can be maintained and promptly managed.

3.1. Classification of Monitoring Objectives

The information provided by nodes and sections of drainage networks in different studies has different meanings and values [6]. For example, to address the information collection needs of a drainage network, the small-scale network in Example 8 of the SWMM (the Storm Water Management Model, which is a comprehensive tool used to simulate urban stormwater runoff and drainage system behavior, widely employed in planning, analysis, and design of stormwater management systems) application manual has been used in several studies as the object. Based on this, several monitoring schemes using different sensor layouts have been developed depending on the objectives [31–33]. This included the monitoring of the operational status and the identification of abnormal conditions in the network.

3.1.1. Monitoring of the Operational Status of the Pipeline Network

The assessment and diagnosis of the pipeline network operation states are mainly realized via the real-time monitoring of hydraulic parameters such as flow rate, liquid level, and flow velocity [34]. By analyzing abnormal signals and sharing data across nodes, IoT sensors enable spatial continuity in monitoring, improving the detection and resolution of pipeline network anomalies. The IoT sensors are optimally arranged. Optimal IoT layouts are used to improve the coverage area [35,36], the amount of monitored information, the time of detection [33,37], and the reliability of the monitoring scheme [30,38].

In addition, the monitored data of the drainage network can be used in the calibration of numerical models. The hydraulic and water quality coupling model can be used to solve hydrodynamic problems involving flow velocity, flow rate, and water level. They can also simulate source diffusion and its inverse [39,40]. However, they require more input parameters [41] for calibration. Both the number and spatial location of calibration points can affect the calibration accuracy of the model [42,43]. By optimizing the arrangement of sensors to collect data at the optimal calibration points [44] of the model, the cost of data collection can be reduced while improving the accuracy.

3.1.2. Identification of Abnormal States in Pipeline Networks

(1) Tracking and Source Tracing of Target Pollutants

By monitoring the composition of sewage such as pH, conductivity [2,4], and other water quality parameters in the drainage network, the tracking and source tracing of illegal discharge in the network can be realized. Based on reverse modeling [45] and the inversion [46] of monitoring data on the transfer and diffusion process of target pollutants, or the establishment of transfer and transport models for these pollutants [20], tracking and source tracing can be achieved. Wastewater treatment plants can be instructed to implement emergency measures in advance to minimize the impact on the treatment process owing to fluctuations in sewage quality in the network.

The drainage network is an important type of infrastructure in cities. Downstream inspection wells collect and reflect the upstream sewage composition information. By monitoring the specific chemical components of wastewater, it is possible to locate and trace target pollutants. In recent years, wastewater-based epidemiology (WBE) has facilitated the tracking and localization of mass epidemic outbreak hotspots [47,48] and illicit drug abuse [49,50]. It has achieved reliable monitoring results in populated areas such as communities [51] and universities [52]. In particular, this method is low cost and allows for wide coverage during the initial screening of large populations compared to the sampling and

testing of individuals. An example of this is during the large-scale COVID-19 outbreak [53]. When information about the target is detected in the drainage network, progressive screening of the upstream area can facilitate the localization of infected individuals. Appropriate isolation measures can then be implemented.

(2) Hotspot Monitoring of Floods and Overflows

The monitoring of water levels in municipal drainage networks via the placement of IoT sensors in flood-prone areas can guide flood emergency response and management, such as in unattended overflow sections of the pipeline network [45], flooding hotspots in the combined drainage network [22], and areas where backflow occurs in rainwater networks during high tides [54]. Through the appropriate positioning of sensors, the scope of investigation can be narrowed for the required measurement. The development of AI technology enables the efficient application of the massive amount of data generation from IoT sensors to flood modeling and prediction. Zhang et al. [55] constructed different neural network models to simulate and predict CSO water levels based on IoT-monitored CSO water-level datasets and rainfall intensity data. C. Cosco et al. [9] considered the three factors of CSO frequency, volume of water, and the amount of pollution, to monitor CSO events in a combined drainage network to minimize their impacts.

3.2. Assumptions for Scheme Design

Before optimizing the layout of IoT sensors, preliminary scheme design is conducted based on premises and assumptions, considering the monitoring objective, drainage network characteristics, and design experience. This can address the information deficit of the design process, anticipate potential problems and risks in advance, and, thus, develop coping strategies in advance.

3.2.1. Risk and Necessity Assumptions for Nodes

In theory, each node and pipeline can be selected as a monitoring point. However, such a large-scale arrangement of monitoring points is not economical. Based on the monitoring objectives, it may be necessary to evaluate the risks and necessity of nodes in different locations to eliminate extraneous nodes and their connecting pipeline segments. There are two main assumptions regarding the necessity of nodes. One is that any node in the pipeline network has an equal probability of being the source in pollutant monitoring. Thus, the nodes have the same weight in the scheme formulation. The other is that the nodes in certain locations in the drainage network have a higher necessity for monitoring. Table 2 lists the assumptions about the risk and necessity of nodal monitoring of UDNs.

Exploiting the idea of Bayesian inference, when arranging sensors, empirical knowledge is used as prior information based on established objectives to classify nodes and assign weights in advance, thereby reducing the number of decision variables and the computational burden during the early stages of establishing the IoT sensor layout optimization model. This is applicable to solving black-box optimization problems, including sensor position selection. Moreover, the results indicate that when only specific monitoring objectives are considered, the probability distribution of prior information does not affect the optimized placement of the sensors, and the optimized result is insensitive to this parameter [13,32]. Therefore, detailed experience and data analysis are not required to create prior information.

Table 2. List of monitoring necessities associated with different types of nodes.

Category	Type of Study	Type of Node
Nodes in certain locations have a higher necessity for monitoring	WEB-based drainage network monitoring	Priority is given to areas with a high population density such as hospitals, nursing homes, community centers, and inspection wells connected to buildings [12,56]. The location of potentially infected nodes is reflected through Bayesian prior probabilities [13]
	Calibration of hydrodynamic models	The pipeline where the flow rate changes is the “potential measurement location” [57]
	Tracking and source tracing of pollutants	Most pollution events occur in nodes located downstream [32,58] Different levels of priority are assigned to each potential monitoring site considering CSO frequency, flow rate, and mass of pollutant [9] A pre-screening procedure based on the concept of pollution matrix is introduced to screen some nodes [59]
	Daily monitoring	Nodes in key topological locations, nodes located in areas with high residential density [31]
Nodes have the same necessity for monitoring [17,33,35]	Monitoring and source tracing of pollutants in drainage networks	Any node has an equal probability of being the source

3.2.2. Assumptions for Sensor Installation and Maintenance

In theory, the installation and operational costs, failure conditions, and service life of the sensors do not change significantly with the location of installation, and the sensors are assumed to be immune to failure and damage. However, in practical situations, sensor failure and delayed response can have an impact on monitoring, especially in water supply networks [60,61]. The installation and maintenance costs of sensors are linked to their specific location [62], and the actual situation must be considered to guide the installation process. Moreover, the sensors are more susceptible to damage owing to complex sewage composition and the operating environment [8]. Therefore, the operational costs, robustness, and adaptability of the design scheme should be comprehensively considered during the scheme design phase.

3.3. Optimization of Monitoring Point Locations

3.3.1. Design Evaluation Methods Based on Engineering Experience

Decision makers need to comprehensively consider the overall study area and objectives when selecting the location of monitoring points based on engineering experience. The general principles considered include the principle of controllability, as well as the requirements for uniformity, feasibility, representativeness, safety, and installation and maintenance costs [5,6,10,25]. Important nodes in the drainage network [12,63], other than inspection wells, or the selection of important drainage facilities, such as sewage treatment plants [64], discharge outlets [24,65], pump stations [6], and weirs [26], all require special attention. In addition, in some cases, where monitoring points are arranged in a UDN [34,66], the classification of branch pipelines, main sewers, and main drains is conducted based

on several factors. These include the connection structure of the network, the location of monitoring points, and monitored parameters. They are determined step by step, and the monitoring points are classified. Ogie et al. [67] considered the high-impact scenarios of sensor monitoring deployment, the high-risk locations of the monitored objects, and the high coverage of the sensor network as the objectives. They set different weights for the objectives based on expert opinions and optimized the objective function to calculate the sensor layout scheme.

In certain UDNs with a long service life, detailed basic data and relevant information are often missing, making it difficult to establish high-precision models. Therefore, expert experience and engineering practice are relied on to select monitoring point locations, and extensive data and information on the operation of drainage networks are acquired. However, for UDNs with complex topology and a large number of pipelines and nodes, it is difficult to directly determine the monitoring point locations based on human experience. For example, the collected data at the discharge outlet and the downstream area can reflect the summary information of the system and facilitate useful observational utility [56,67]. However, the merging of different branch pipelines blurs the details of the spatial information [57]. Thus, the layout of monitoring points should be more scientific and reasonable.

3.3.2. Design Evaluation Methods Based on Information Theory

Information theory can also be used for the design evaluation of optimized sensor layouts. Entropy is defined as a reduction in the uncertainty of a random variable X [68] in information theory. When a signal is observed, the amount of information obtained indirectly measures the reduction in uncertainty. To some extent, the concepts of information and uncertainty can be equated [69]. Information entropy and some of its related concepts, as shown in Table 3, have been applied to the design and evaluation of a variety of hydrological measurement networks, including precipitation [70,71], runoff [72–74], water quality [69], groundwater [21,75], polder [26], and water supply networks [6].

Table 3. Formulas and meanings of entropy and its related concepts [76].

Entropy-Based Information Measure	Calculation Formula	Meaning
Entropy (high uncertainty entropy, information entropy or marginal entropy)	$H(X) = - \sum_{k=1}^K p(x_k) \log[p(x_k)]$	Amount of information provided by each monitoring point
Joint entropy (joint information entropy, total entropy, multivariate joint entropy)	$H(X_1, X_2, \dots, X_N) = - \sum_{x_1 \in X_1} \sum_{x_2 \in X_2} \dots \sum_{x_n \in X_n} p(x_1, x_2, \dots, x_n) \log_2 p(x_1, x_2, \dots, x_n)$	Total amount of information provided by multiple monitoring points together
Total correlation	$C(X_1, X_2, \dots, X_N) = \sum_{i=1}^N H(X_i) - H(X_1, X_2, \dots, X_N)$	Redundancy of information between variables
Transinformation (mutual information) The special cases of total correlation, where $n = 2$ [26]	$I(X_1, X_2) = H(X_1) + H(X_2) - H(X_1, X_2)$	Components of redundant information in the monitoring network
Conditional entropy	$H(X_1 X_2) = H(X_1, X_2) - H(X_2)$	the information loss that occurs during the trans-information process between random variates X_1 and X_2

Li et al. [76] proposed a generalized maximum information minimum redundancy (MIMR) criterion for the design and evaluation of hydrological networks, which satisfies the three objectives of maximum overall information (joint entropy), maximum information transformation capability (ubiquitous information), and minimum redundancy information (total correlation). The design and evaluation of monitoring schemes based on the MIMR

criterion can accurately and efficiently represent information. This criterion has been further supplemented and improved by researchers in the field. Combining heuristics, such as greedy algorithms, genetic algorithms, and evolutionary algorithms, it has also been widely applied to the optimized layout of IoT sensors in UDNs and hydrological network monitoring [1,27].

Based on the principle of information theory, high entropy corresponds to high uncertainty (and high information content) in measurements owing to a large number of possible events [1]. Therefore, locations with high entropy values [37,77] are usually selected as monitoring points. However, points with high information content may not necessarily have high monitoring value. Alfonso and Price [28] introduced the concept of information value when designing hydrological monitoring networks. Different decisions were made based on the information acquired from monitoring points. The resulting consequences corresponded to different information values, and locations with high information values [6,78] served as monitoring points. This method can guide the decision-making process regarding the operational status of water conservancy facilities. A major drawback of entropy-based criteria is that nodes with low redundant information are usually located as far away from each other as possible [77]. As a result, these schemes tend to place sensors on the boundaries of the study area, thus ignoring practical engineering needs. Moreover, population density and socio-geographical issues are not considered [79].

3.3.3. Design Evaluation Methods Based on Statistical Theory

The main idea behind the design and optimization of the monitoring network is to reduce the uncertainty associated with unmonitored locations [26], as well as the redundancy of monitored locations. Statistical theory can handle and analyze massive amounts of data as well as measure and quantify uncertainty. Therefore, statistical theory represented by cluster analysis is widely used in the design of sensor locations for water supply [80–83]. Cluster analysis groups unlabeled data points into clusters based on similarity. This method also categorizes nodes using the similarity of data patterns or topological relationships. Thus, it is possible to reduce the number of monitoring points while ensuring overall effectiveness, thereby guiding the selection of sensor locations.

However, the number of required clusters for analysis must be determined by decision makers after several tests [22,31] or by using heuristic algorithms [84]. It may not be possible to obtain the optimal cluster partitioning, and the results cannot identify specific sensor location choices within the cluster. Given that the focus is on the similarity of nodes, the benefits and value of single monitoring points are ignored.

3.3.4. Design Evaluation Methods Based on Complex Network Theory

Most optimized IoT sensor arrangements are designed based on the time-series datasets of monitoring points [22] or pollutant transport and dispersion distributions in the pipeline network [35,39,58] based on SWMM pipeline network hydrodynamic simulations. Other methods have also been adopted instead of hydrodynamic models, focusing on the topological structure of drainage networks. The placement of sensors was optimized by evaluating the degree to which each node was affected by abnormal events in the pipeline network. Simone et al. [85] analyzed the potential and feasibility of complex network theory (CNT) in the assessment of the vulnerability and resilience of UDNs, the design of optimal monitoring points, and the study of the propagation of pollutants. They also proposed a series of topology-based indexes and coefficients that were utilized to guide the optimal placement of sensors. This approach considers the relative position and connections between the nodes and facilitates the optimal arrangement and evaluation

of the sensors without performing hydraulic simulations, which serves as an alternative approach to solving these types of problems.

3.3.5. Design Evaluation Methods Based on Observability Theory

The observability of a complex system can be defined as the utilization of the minimum number of sensors to collect enough information for the prediction or estimation of the state of areas of the system that are not directly monitored [86]. This facilitates a better understanding of the system's operational status and improves situational awareness for risk management and decision making. Observability analysis can be used to establish a state space model to ensure that the system's state can be estimated using sensor measurements. The approach guides the location selection process of sensors in applications such as the evaluation of the pollutant concentration in river networks [11] and the identification of key nodes in flood monitoring networks [87]. In the water supply network, the system's state can be estimated based on indicators such as node water demand [88], disinfectant (chlorine) concentration [89], and pollutant identification [90], and sensors can then be arranged accordingly. There are few studies on the design and evaluation of the optimized placement of IoT sensors in UDNs based on observability. Zheng et al. [78] used the matrix completion model to improve observability by recovering missing data from unmonitored locations. They quantified the deviation between the predicted value of the water level and the observed value to evaluate the optimized placement scheme of the sensors. The utilization of the minimum number of optimally arranged sensors in a complex network structure to improve the observability of the pipeline network has great application potential for the identification and early warning of abnormal states in the operation of a UDN. In Table 4, we summarize and compare the properties of the above methods in terms of accuracy, robustness, computational costs, and adaptability.

Table 4. Performance comparison of main optimized methods.

Method	Accuracy	Robustness	Computational Cost	Adaptability
Cluster analysis	Medium, dependent on data distribution and algorithm selection	low, parameters need to be preset, sensitive to noise	Moderate algorithmic complexity	Ideal for rapid deployment and cost control
Information theory	High, quantifying information content through information entropy	Medium, relies on data distribution stability, performance degrades with data noise or dynamic changes	High, requires complex optimization algorithms	Not applicable to old pipe networks or where data is missing
Complex network theory	Medium, based on topology, easy to overlook hydraulic details	High, topologically stable	Low, only network analysis required, no complex simulations	Not dependent on hydraulic parameters, but accuracy is limited
Observability theory	High, inferring state through mathematical modeling	High, model parameters optimized for noise immunity	High, involving matrix operations	Effectiveness is limited when data is extremely sparse

3.3.6. Hybrid Optimization Methods and Application

In this section, we introduce the application of the above optimization theories and methods in the practical engineering of the optimal arrangement of IoT sensors in UDNs, including the type of monitoring indicators, monitoring objectives, the size of the pipe network in the monitoring area, the number of monitoring points, and so on. By using the above optimization theories directly or combining them, or combining multiple models and substituting the model outputs into another model for optimization analysis, the advantages of each method can be better exploited to obtain a more effective monitoring scheme. Table 5 summarizes the specific cases of the current IoT optimization arrangement for a UDN.

Table 5. Application of optimal theory and methods in UDN.

Theory	Research	Type		Objectives/ Indicators	Size of Study Network				Monitoring Points	Evaluation Methodol- ogy/Indicators
		Hydraulics	Quality		Area	Conduits/ Pipelines	Nodes/ Junctions	Other Water Facilities		
Multi-objective optimization + information theory	[91]		✓	Maximizing marginal entropy and maximizing trans-information (all entropy variates have positive values)	161 ha	\	80	\	The outlet + other 7 monitor- ing points	
	[3]		✓	Monitoring of pollution of water quality in the sewer network	12 sub-catchments, covering an area of 19.71 km ²	1909	1902	14 pumps, 14 storage units and 1 treat- ment plant	12	Scheme with maximized joint entropy and minimized total correlation
	[37]		✓		12 sub-catchments, covering an area of 19.71 km ²	1909	1902	14 pumps, 14 storage units and 1 treat- ment plant	14	
	[79]		✓		4 separate catchment areas	\	748 candi- date sites	\	20	
	[78]	✓		understanding the operating status of UDS and facilitating urban flood early warning.	2.679 km ² , is divided into 2693 sub- catchments	898	878	\	4 monitoring points (lower budget) or 8 monitoring points (higher budget)	Maximize value of information (VOI), minimize trans-information (TE) and minimize economic costs

Table 5. Cont.

Theory	Research	Type		Objectives/ Indicators	Size of Study Network				Monitoring Points	Evaluation Methodol- ogy/Indicators
		Hydraulics	Quality		Area	Conduits/ Pipelines	Nodes/ Junctions	Other Water Facilities		
Complex network theory (CNT)	[36]		✓	Evaluating the ability of node to receive pollutants, detecting the maximum amount of information propagated on the network	\	79	77	1 outfall	4	In-Relevance-Harmonic Centrality as the indicator
	[85]		✓	Analyzing the impact of a pollutant spill at a given node on the entire system, focusing on the role of network topology in pollutant dispersion	SWMM example 3	32	32	1 outfall, 1 storage and 1 pump.	\	node contamination index IC as the indicator
	[35]		✓	Analyzing the dispersion of pollutants by calculating the influence coefficients of each node with respect to the installed sensor in the system (usually located at the outfall).	\	79	77	1 outfall	7 monitoring points, with a network coverage equal to 60%	impact coefficient (IC) as the indicator
	[38]		✓	Analyzing pollutant dispersion at various nodes	\	79	77	1 outfall	7 monitoring points and the prioritization order of the sensors was further considered	topological impact coefficient (IC ^T) as the indicator

Table 5. Cont.

Theory	Research	Type		Objectives/ Indicators	Size of Study Network				Monitoring Points	Evaluation Methodol- ogy/Indicators
		Hydraulics	Quality		Area	Conduits/ Pipelines	Nodes/ Junctions	Other Water Facilities		
Multi-objective optimization + Expert advice	[67]	✓		Obtaining water level data to make decisions about the operation of flood control infrastructure	\	647 water- ways	579	71 pumping sta- tions, 30 flood- gates and 11 flood gauges	4 sensors in the first phase, which will be expanded to 10 sensors in the follow- ing phases	Locations at high risk of monitored phenomena; loca- tions to maxi- mize network coverage; water level information to maximize water temperature infrastructure
	[92]	✓	✓		\	4 main pipelines	\	\	After optimization 10 monitoring points can replace all the initial monitoring points	Statistical significance analyses based on F-tests and t-tests were performed to compare the data for differences between groups
	[93]	✓	✓	Improving monitoring of operational status in sever network	a catchment	\	\	\	The original 23 monitoring sites were reduced to 10 sites through optimization	
Cluster analysis	[30]	✓			21.5 km ²	192	187	17 outfalls	12	The composite indicator compounded from Pearson’s coefficient and Euclidean distance

Table 5. Cont.

Theory	Research	Type		Objectives/ Indicators	Size of Study Network				Monitoring Points	Evaluation Methodol- ogy/Indicators
		Hydraulics	Quality		Area	Conduits/ Pipelines	Nodes/ Junctions	Other Water Facilities		
Cluster analysis	[84]	✓	✓	Improving monitoring of operational status in sewer network	62.88 km ²	\	23 original monitoring points	\	Reduction of 7 monitor- ing points	Calinski- Harabasz Index, Within-Group Sum of Squares (WGSS), Between-Group Sum of Squares (BGSS)
	[31]	✓	✓		2.63 km ²	785	785	1 outlet	20	Scheme with maximized joint entropy and minimized to- tal correlation
	[22]	✓		Flood hotspot monitoring in combined drainage systems	7 sub-catchments	58	60	2 outfalls	3 or 4	Silhouette Coefficient Index (SCI)
	[20]		✓	Monitoring of pollution of water quality in the sewer network		139	135	\	8	Silhouette Coefficient (SC)
	[94]	✓		detecting inflow and infiltration (I&I) in urban sewer networks	15 km ²	819	820	1 outlet	20	detection reliability (DR); distribution uniformity (DU)

In addition, the optimal layout of sensors was transformed into a pollution traceability problem to improve the accuracy and reliability of source identification (SI) [95,96], and the results obtained were used as a reference for the siting of sensors. Compared with the sensor layout, only by virtue of engineering experience, considering the optimized layout through the above method, the number and distribution of sensors in the UDN are more reasonable, and the performance in the specific evaluation indexes has a better effect; alternatively, the original monitoring layout can be used to ensure the monitoring effect remains unchanged, which reduces the number of sensors and reduces the cost of the layout and maintenance [84,92,93]. However, it is necessary to point out that the theoretical optimization obtained by the layout should also be considered in combination with cost, field layout, and other factors.

3.4. Quantitative Evaluation of the Monitoring Scheme

Optimal IoT layouts are determined by comparing multiple approaches, which fall into two main categories: ① assessing monitoring efficiency through defined metrics, and ② evaluating algorithmic complexity in the optimization process and to quantitatively evaluate the different schemes by the algorithms' performances and related indicators. Evaluation methodology/indicators in Table 5 show some of the quantitative evaluation methods and indicators used by the authors, and in Table 6, we further explain the types and the meaning of evaluation indicators.

Table 6. Quantitative evaluation indexes of optimized IoT sensor layout scheme for drainage network.

Classification	Evaluation Indicator	Meaning
Algorithm performance indicator	Algorithm running time [2,37,42], hypervolume metric (HV) [56], Silhouette coefficient index (SC) [22], Calinski-Harabaz index (CH) [84].	Quantitative evaluation of optimized sensor placement schemes via evaluation of algorithm performance
	Mean Absolute Error (MAE), Root Mean Square Error (RMSE), relative RMSE (RRE), coefficient of determination (R^2), Nash-Sutcliffe Efficiency (NSE) [20,56,57], Confidence Coefficient, Segregation Likelihood [58], Residual Squared, Correlation Coefficient, Consistency Index [57], Mean of Relative Error MRE [46].	Differences between predicted and actual values obtained by inversion when studying the transport and traceability of pollutants
Assessment of the achievement of objectives	Number of detected events and event detection rate [39,58], access to monitoring information wherever possible [31,33,84], response time to anomalous events [20,37], reliability of the monitoring network [30,33,37,58,94] uniformity of sensor distribution [94].	Assessed by the degree of achievement of monitoring objectives

Due to the diversity of monitoring objectives, optimization theories, and methods, the quantitative evaluation of the optimal deployment of IoT sensors in UDNs has not yet formed a unified evaluation method and process and even lacks the step of quantitatively evaluating the scheme in some studies. On the basis of the existing research results and with reference to the evaluation of other sensor optimization schemes, the performance, cost, and feasibility of the implementation of the optimization scheme of the sensor design scheme are comprehensively evaluated through the quantification of indicators of different dimensions, including the following: for the formulated monitoring objectives, the collected data can effectively respond to the operational status of the UDN; the arrangement of the sensors is cost-effective, and the scope covers the study area while avoiding data redundancy.

4. Summary and Recommendations

This review focuses on a strategy for the optimal layout of IoT sensors for UDNs, covering the general process, principal methods, and application cases. The main conclusions are listed as follows:

- (1) The monitoring of drainage networks primarily reflects their operational status and detects and identifies abnormal states through hydraulic and water quality parameters. Different monitoring areas correspond to distinct monitoring indexes, which, in turn, gives rise to varying layout schemes. Nevertheless, the arrangement of IoT sensors for different networks should strive to provide the maximum amount of valid information.
- (2) In practical scenarios, multiple types of sensors are required for collaborative monitoring and sensing of UDNs. The sensor arrangement methods, based on information theory, statistical theory, and complex network theory, have their own characteristics and adaptation of application. Modularly combining these methods or employing hybrid optimal strategies can be useful for the design and evaluation of optimal layout solutions.
- (3) The quantitative evaluation of monitoring schemes is mainly conducted based on the degree of objective achievement, performance evaluation indexes of optimization algorithms, comparisons with the original sensor layout scheme, or contrasts in the effects achieved by different optimization methods. Nevertheless, a systematic and standardized quantitative evaluation process and methodology for monitoring schemes remain to be established. The current optimized layout methods may not be optimal and deserve further study.
- (4) The existing research on the optimal layout of sensors predominantly focuses on spatial distribution and quantity. In the future, dynamic optimization strategies can be explored by implementing a phased deployment–feedback iteration approach. In the initial stage, sensors should be preferentially arranged at critical nodes. Subsequently, the arrangement scheme for the next stage can be adjusted and determined according to the collected feedback data and information gain. The dynamic optimization strategy has the potential to enhance the accuracy and objectivity of scheme design and evaluation.

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